

Structured Local Adaptation and Hybrid Adjudication in Ultra-Low-Resource Tangut Short-Text Translation

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Abstract

Tangut translation is an extreme low-resource problem: the current repository contains only 491 real Tangut-Chinese pairs, split 400/41/50 into train/dev/test, and many targets are compact title-like Chinese strings rather than ordinary sentences. We compare local prompt-only baselines, a stronger frontier prompt-only baseline, synthetic-data supervised fine-tuning (SFT), direct preference optimization (DPO), and inference-time hybrid adjudication. DeepSeek-V3.2 with dictionary grounding and four fixed few-shot exemplars is the strongest single model, reaching 2.80/5 on a reference-aware judge with 12/50 exact matches. Within the open/local regime, structured multitask SFT is the best supervised base, and DPO helps only on that base with strict preference-gap filtering: the gap-0.4 sigmoid variant reaches 30.01 chrF++ and is the strongest local model. A catalog-aware adjudicator over the frontier output, UNK-SFT, and that DPO model yields the strongest non-reference system overall, with 34.87 chrF++, 3.02 reference-aware overall, 14/50 exact matches, and zero contamination. Bootstrap resampling confirms a robust chrF++ gain over the frontier baseline (+6.33, 95% CI [2.10, 11.67]); judge and exact-match gains are positive but boundary-level on this 50-item test set. A local Qwen judge preserves the same frontier < hybrid < human ordering, and a deterministic open selector over the same pool still reaches 32.62 chrF++, showing that part of the workflow gain is reproducible without closed adjudication. Finally, lexical coverage and perplexity mis-rank the gold reference itself, so reference-aware evidence should be treated as primary and reference-free metrics as diagnostic. The practical lesson is two-level: strong frontier prompting is the best single-model upper bound, but structured local training remains useful because it supplies complementary historical-title hypotheses for a stronger scholar-facing workflow.

1 Introduction

Machine translation for historical ultra-low-resource languages is hard for two separate reasons: parallel data are scarce, and the outputs that matter to scholars are often short titles or gloss-like strings rather than fluent modern sentences. Recent work on ancient-language NLP and historical-text translation has made this setting more visible, but it has also shown that both supervision and evaluation become unstable outside modern benchmark conditions [1, 2, 3]. Tangut is a canonical example. In the current repository snapshot, only 491 real Tangut-Chinese pairs are available.

This regime also creates an obvious objection: perhaps a strong Chinese frontier LLM, given dictionary glosses and a careful title-oriented prompt, is already enough. We therefore compare five components in one frame: local dictionary-grounded prompting, a stronger proprietary frontier prompt-only baseline, synthetic pseudo-parallel SFT, preference optimization on top of SFT, and a new inference-time hybrid adjudication workflow [5, 4, 11]. The point is to separate what prompting already buys, what local training still buys, and whether the two can be combined into a stronger scholar-facing system.

Our findings are straightforward. The stronger prompt-only objection is real: the DeepSeek-V3.2 few-shot dictionary baseline is the strongest single model we test, with 2.80/5 overall under the

reference-aware judge and 12/50 exact matches. Within the open/local trainable regime, structured multitask SFT is the right base, and DPO helps only after it is moved onto that base and the preference pairs are filtered much more aggressively; the strict gap-filtered sigmoid variant reaches the highest local chrF++ in the repository (30.01). Complementarity is also not merely an oracle observation. A catalog-aware three-way adjudicator over the frontier output, UNK-SFT, and the strict gap-filtered DPO model reaches 34.87 chrF++, 3.02 overall, and 14 exact matches, while a deterministic open selector over the same pool still recovers part of that gain without any closed adjudicator. Finally, the repository exposes two robustness issues directly: several reference-free metrics mis-rank the gold reference itself, and low-gap preference data remain noisy enough to destabilize DPO.

The current draft therefore makes three contributions:

1. an empirical comparison that cleanly separates the strongest single model, the strongest local trainable recipe, and the strongest hybrid workflow under only 491 real pairs;
2. a positive local-method result showing that structured multitask SFT is the right base and that strict gap-filtered DPO produces the strongest local model, while catalog-aware hybrid adjudication turns those local candidates into a better workflow and a deterministic open selector recovers part of the same gain;
3. a reference-aware evaluation, robustness, and data-hygiene analysis showing why reference-free metrics and low-gap preference data can mislead in this task.

The rest of the paper situates the task against related work, describes the comparison families, and then keeps two claims separate: how to build the best open Tangut model, and how to combine frontier and local candidates into a stronger workflow.

2 Related Work

Ancient and historical language processing. Ancient-language NLP differs from mainstream NLP not only because data are scarce, but also because scripts, editorial conventions, and annotation practices are unstable across corpora [1]. Shared tasks on historical translation and document modernization report similar sparsity, though they often target broader sentence translation or orthographic updating rather than cross-script title reconstruction [2, 3]. Our Tangut setting is narrower still: only 491 real pairs are available, and the evaluation set is dominated by compact title-like strings.

Low-resource MT, synthetic supervision, and lexical grounding. Low-resource MT often relies on synthetic parallel data, multilingual transfer, or explicit lexical constraints [4, 6, 8, 5, 7]. Tangut overlaps with this literature but is harsher in two ways: the source script is largely outside standard MT pretraining, and the outputs are compact catalog titles rather than ordinary sentences. We therefore focus on synthetic supervision and lexicon-grounded structure rather than on broad multilingual transfer.

Dictionary-structured prompting and error-aware prompting. Recent LLM translation work suggests that lexical resources are most useful when presented as structured intermediate evidence rather than as flat prompt context. Chain-of-Dictionary prompting and Learning-From-Mistakes prompting both follow this pattern in low-resource settings [9, 10]. Our stronger frontier baseline is adjacent to this line, but the present paper asks a different question: whether alignment

structure should be absorbed into trainable local targets, and whether those local hypotheses remain useful after a strong frontier prompt-only baseline is already in place.

Preference optimization. Direct Preference Optimization (DPO) is now a standard offline alignment objective, and recent work has extended it to machine translation and noisy-preference settings [11, 14, 15, 12, 13]. Our results place this literature in a harsher regime: supervision is tiny, outputs are title-like, and the chosen/rejected signal is built from partially misaligned automatic metrics. In this setting, DPO helps only after base-model choice and pair filtering are controlled aggressively.

Evaluation for non-standard translation outputs. chrF++ is widely used in machine translation because character-level overlap is often more stable than token-level overlap under orthographic variation [16]. Learned MT metrics and QE systems such as COMET and OpenKiwi go beyond surface overlap, but recent work also shows that they can have domain-specific failure modes and induce metric interference [17, 20, 18, 19]. Our findings sharpen that caution in a historical short-text regime: lexical coverage and perplexity can penalize the gold reference itself, so reference-aware evidence must remain primary.

3 Task Setting

The real Tangut data in this repository contains 491 pairs. The current split is 400 training examples, 41 development examples, and 50 test examples, produced by a single random shuffle with seed 42. Unlike ordinary machine-translation benchmarks, the targets are short and often title-like. Many examples resemble book titles, sutra titles, or compact scholastic expressions. This matters because title reconstruction rewards precise lexical choices and compact formatting, while a modern Chinese fluency model may incorrectly treat those outputs as unnatural.

Because the real dataset is tiny, the system uses synthetic data to train translation models. For each SFT branch, 50,000 synthetic Tangut-Chinese pairs are produced from monolingual classical Chinese and dictionary knowledge, then combined with the 400 real training examples upsampled by a factor of 10 to form a 54,000-example training set. All learned models use the same instruction-tuned 7B backbone. These synthetic inputs are pseudo-Tangut training proxies rather than claims about faithful Tangut reconstruction; in other words, we judge them only by downstream utility on held-out real Tangut titles, not by intrinsic linguistic fidelity.

We also explicitly audit data hygiene. There is zero exact source, target, or full-pair overlap across the train/dev/test splits, zero exact overlap between the 50 test references and the final modern-Chinese targets of the 50k synthetic multitask corpus, and zero exact overlap between the four development examples used in the frontier few-shot prompt and the test references. Character n -gram overlap is naturally non-zero inside the real dataset because titles share common Buddhist morphemes, but the synthetic multitask targets are far from copied test references: their mean maximum Jaccard overlap with test references is 0.027 for character bigrams, 0.004 for trigrams, and 0.000 for 4-grams.

We evaluate with two classes of metrics. The repository’s existing metrics are lexical coverage, perplexity, chrF++, and a dictionary-conditioned LLM judge. chrF++ is a reasonable overlap anchor for this setting because character-based scoring is relatively stable under orthographic variation [16]. Perplexity is computed as $\exp(\ell)$, where ℓ is the mean token-level causal-LM loss under Qwen2.5-0.5B. In this title-heavy setting that score becomes extremely large even for the gold reference, so we treat it as diagnostic rather than dispositive. “Contamination” in the current

paper likewise needs a narrow definition: it means mixed-script or artifact contamination in the *output* string, such as Latin letters, literal [UNK] tokens, or prompt residue, not train-test data leakage. Learned MT evaluation and quality-estimation systems can capture richer signals than surface overlap [17, 20], but recent work also shows that strong neural MT metrics can have hidden domain-specific failure modes and can induce metric interference when similar signals appear in both system development and final evaluation [18, 19]. For that reason, we do not use off-the-shelf COMET as a headline arbiter in this draft: standard checkpoints are designed for modern sentence-level MT rather than Tangut-script short-title reconstruction, so any COMET score would still require task-specific validation. Accordingly, this draft does not treat all metrics equally: chrF++, exact match, contamination, suffix preservation, and the new reference-aware judge are primary evidence, while lexical coverage, perplexity, and the legacy reference-free judge are reported as diagnostics.

4 Methods

4.1 Inference-only baselines

The first family contains no weight updates. **baseline1** prompts the local model directly on the Tangut string; **baseline2** adds dictionary glosses; and **baseline2_1_cot** further decomposes generation into alignment, literal drafting, and rewriting. This family is closest to lexicon-augmented MT, except that the lexicon is injected through prompting rather than decoder constraints [5, 7].

We also add a deliberately stronger prompt-only comparison based on DeepSeek-V3.2 with dictionary grounding and four fixed development-set exemplars. Its prompt emphasizes compact title form, discourages explanatory paraphrase, and emits only the final title string. Decoding uses temperature 0.0 with a 512-token cap. The goal is not to build a straw baseline but to test directly whether frontier prompting already solves the task well enough.

4.2 Synthetic SFT baselines

The second family uses synthetic training data. We build 50k pseudo-Tangut/Chinese pairs from monolingual historical Chinese by mapping Chinese characters to Tangut characters through dictionary explanations, fixing the synthesis seed to 42, and keeping source lengths between 4 and 64 characters. The four branches differ only in how dictionary misses are handled and whether the target includes alignment structure: mixed-script keeps 30–70% Chinese characters, UNK replaces misses with [UNK], semantic projection replaces misses through Qwen2.5-0.5B embeddings plus FAISS lookup, and multitask keeps the UNK input but changes the target to `<dict_match>{...}</dict_match><literal>{...}</literal> {final title}`. This gives the model both an alignment channel and a final title target in one sequence. These examples are not philologically faithful Tangut reconstructions; they are structured training proxies. Each synthetic set is combined with the 400 real training examples upsampled by a factor of 10 and shuffled with seed 42. The key design question is whether the model benefits more from input purity, approximate semantic preservation, or explicit alignment structure.

4.3 Preference optimization

The final family applies DPO to an SFT base using reward-defined chosen/rejected pairs [11]. We sample five outputs per training prompt at temperature 0.8 and top- p 0.95, then score them with

$$r(y) = \text{lex}(y) - 0.01 \log \text{ppl}(y),$$

where $\text{lex}(y)$ is the repository’s lexical-coverage score. Concretely, lexical coverage segments the Chinese output with jieba plus single-character backoff, performs maximum forward matching over the Tangut input with dictionary entries up to length five, and reports the fraction of matched Tangut spans whose gloss set intersects the output token set. The $\log \text{ppl}(y)$ term is down-weighted by 0.01 so that fluency remains secondary. We then keep the top-scoring output as chosen and the bottom-scoring output as rejected whenever the reward gap is at least 0.05.

The legacy pipeline first selects the best SFT base among the mixed, UNK, and semantic variants using `chrF++`, then constructs preference pairs from that base. In the current snapshot that selector chooses the UNK model, yielding the legacy checkpoint `final_v2`, which serves as a negative control. The follow-up DPO runs instead start from the stronger multitask SFT model. After auditing the preference data, we keep only pairs whose reward gap exceeds either 0.2 (1,996 pairs) or 0.4 (498 pairs), and we train both standard sigmoid DPO and a robust+weighted variant [12, 13]. This turns the DPO family into a controlled test of base-model choice, pair filtering, and loss robustness.

4.4 Catalog-aware hybrid adjudication

The oracle analysis suggested that frontier and local candidates are complementary, but an oracle is not a deployable system. We therefore build a real inference-time adjudicator over the frontier DeepSeek baseline, UNK-SFT, and MT-DPO g0.4 sig. For each item it sees the Tangut source, dictionary glosses, the three candidate titles, their lengths, and simple prefix/suffix consensus hints.

The adjudicator uses Azure-hosted GPT-5.4 at temperature 0.0. Its prompt treats the task as historical catalog-title reconstruction rather than free paraphrase: dictionary glosses are clues rather than canonical modern titles, and short inputs are penalized for explanatory expansion. The model may either copy one candidate or perform a minimal repair using only candidate fragments or directly attested gloss segments. The repository stores the exact templates, frontier prompting script, fixed exemplar IDs, and per-example adjudication JSON, so the closed-model part of the workflow is at least auditable. For reproducibility, we also add a deterministic open reranker over the same pool. It never synthesizes text or calls an external model; it selects one candidate using prefix/suffix consensus, title-suffix priors, anti-truncation and anti-expansion guards, contamination penalties, and a small frontier prior.

4.5 Training and judge details

All learned models start from Qwen2.5-7B-Instruct with LoRA adapters. SFT uses three epochs, learning rate 2×10^{-4} , batch size 4, gradient accumulation 4, rank-64 LoRA, and maximum sequence length 512. DPO uses two epochs, learning rate 5×10^{-5} , batch size 1, gradient accumulation 8, $\beta = 0.1$, rank-32 LoRA, cosine decay, and the same sequence length; the robust variants add label smoothing 0.1 and WPO-style weighting. The full pipeline fixes seed 42 for synthesis, data combination, and DPO training.

The reference-aware judge is a supplementary evaluator, not a training signal. It uses Azure-hosted GPT-5.4 to score reference agreement, source faithfulness, title-style fitness, and overall quality on a 1–5 scale after seeing the Tangut source, gold reference, and candidate simultaneously. Final scores are simple means over examples. For cross-judge robustness, the repository also includes a local Qwen2.5-7B-Instruct judge with the same rubric. We use the Azure judge as the primary table because it is the stricter and less saturated scorer.

Method	Lex $\uparrow$	PPL $\downarrow$	chrF++ $\uparrow$	Sem $\uparrow$	Flu $\uparrow$
<i>Prompt-only</i>					
Zero-shot	0.1165	20.21	0.19	1.00	1.56
Dict prompt	0.8187	4474.44	6.13	2.30	2.40
Dict+CoT	0.8723	18119.16	7.59	2.30	1.88
<i>Frontier DeepSeek</i>	0.6234	7438.72	28.54	2.62	2.78
<i>Local learned</i>					
Mixed-SFT	0.3192	298684.91	17.85	1.42	1.88
UNK-SFT	0.3514	11031.18	22.11	1.66	1.96
MT-SFT	0.4839	3322.93	25.99	1.70	2.26
Sem-SFT	0.3723	17064.38	17.03	1.46	1.84
Legacy DPO	0.4224	314577.16	19.39	1.52	1.76
MT-DPO g0.2 sig	0.5694	303438.00	20.93	1.88	1.86
MT-DPO g0.2 rob	0.5249	238892.42	27.67	1.88	1.88
MT-DPO g0.4 sig	0.4833	9573.71	30.01	1.84	2.04
MT-DPO g0.4 rob	0.4547	20012.69	27.52	1.74	2.08
<i>Hybrid workflow</i>					
<i>Catalog hybrid</i>	0.6189	6351.88	34.87	2.56	2.70
Human ref.	0.5733	1646139.34	100.00	2.24	2.34

Table 1: Repository-level metrics on the 50-example Tangut test set. Frontier DeepSeek is the strongest single-model baseline. The catalog hybrid adjudicates among the frontier output, UNK-SFT, and MT-DPO g0.4 sig. Here g0.2 and g0.4 denote minimum preference-gap thresholds. Bold marks the best local learned value in each column.

5 Main Results

Table 1 now separates three questions that earlier drafts blurred together: what the strongest prompt-only single model can do, what the strongest open/local trainable model can do, and what a real hybrid workflow can do. The first point is straightforward. The stronger Chinese frontier objection is real: the DeepSeek-V3.2 few-shot dictionary baseline is a genuinely strong single-model result. It reaches 28.54 chrF++, 2.62 semantic completeness, and 2.78 fluency on the legacy repository judge, and later remains the strongest single model under the reference-aware judge.

The positive local-method result still lies inside the trainable regime. Structured multitask SFT remains the strongest pure supervised baseline. Once DPO is rerun on that base with reward-gap filtering, the story changes again: strict filtering does not just rescue DPO qualitatively, it produces the strongest local learned configuration in the repository. The clearest example is MT-DPO g0.4 sig. Tightening the reward-gap threshold from 0.2 to 0.4 cuts the preference set from 1,996 pairs to 498, raises local chrF++ from 20.93 to 30.01, and brings the average output length back near the reference ratio (1.12 instead of 1.74). The strict robust variant is slightly weaker on chrF++ but cleaner than its gap-0.2 counterpart. By contrast, the legacy DPO checkpoint remains a genuine negative control.

The strongest overall non-reference result, however, is no longer a single model. It is the catalog-aware hybrid workflow. Adjudicating among the frontier output, UNK-SFT, and MT-DPO g0.4 sig yields 34.87 chrF++, 14/50 exact matches, and the strongest legacy-judge fluency among non-reference systems in Table 1. This already changes the paper’s practical message. The best question is not “frontier or local?” but “what can local candidates add to a frontier-first workflow if adjudication is conservative and title-aware?”

Not all of that workflow story depends on a closed adjudicator. A deterministic open selector over the same frontier/UNK-SFT/MT-DPO pool, described in the methods section but omitted from Table 1 because it does not use the legacy repository submetrics as primary evidence, still reaches 32.62 chrF++, 12/50 exact matches, and zero contamination. This does not match the full catalog hybrid, but it shows that part of the gain is reproducible with a cheap open reranker rather than only with GPT-based adjudication.

These disagreements are precisely why the paper should not rely on a single automatic metric. Raw chrF++, contamination, and the reference-aware judge each reward different parts of the short-title problem. The defensible claim is therefore two-level rather than absolutist: strict gap-filtered multitask DPO is still the best local training recipe, but catalog-aware adjudication turns that local complementarity into the strongest overall workflow result.

The new statistical analysis makes both layers more precise. Under 5,000 paired bootstrap resamples, the best local DPO model improves over multitask SFT by +4.02 chrF++ with a 95% confidence interval of [0.84, 7.80], so the local chrF++ gain is not just a single-split fluctuation. More importantly for the revised paper, the catalog hybrid improves over the frontier baseline by +6.33 chrF++ with a 95% confidence interval of [2.10, 11.67]. The gains in reference-aware overall (+0.22, 95% CI [0.00, 0.48]) and exact match (+0.04, 95% CI [0.00, 0.10]) are positive but boundary-level under this 50-example test set. This is still a strong workflow result: the hybrid is clearly better on raw overlap, clearly no worse on the reference-aware judge, and directionally better on exact match, while keeping zero contamination and near-reference title length.

6 Analysis

6.1 Reference-aligned diagnostics

Table 2 summarizes the most relevant output-side diagnostics. The frontier DeepSeek baseline is already a clean single model: 12/50 exact matches, 0 contamination, 24/27 suffix-preserving titles, and a near-reference length ratio of 0.95. Within the local family, multitask SFT is the cleanest pure SFT system, and strict gap-0.4 DPO regularizes the outputs further: MT-DPO g0.4 sig restores the length ratio to 1.12 and reaches the highest local chrF++ in the repository. The catalog hybrid is cleaner still, with 14/50 exact matches, 26/27 suffix preservation, 0 contamination, and the same 0.95 length ratio as the frontier baseline. The workflow gain therefore does not come from verbose rewriting; it comes from more accurate conservative title reconstruction.

6.2 From oracle complementarity to a real hybrid

The oracle analysis showed that frontier and local candidates were not near-duplicates. The stronger result is that this complementarity can be operationalized. Table 3 traces the progression from the frontier baseline, to an earlier 2-way hybrid, to the final 3-way catalog hybrid, and then to the best-of-three oracle.

The 3-way catalog hybrid changes 13/50 frontier predictions, reaches 34.87 chrF++, 3.02 reference-aware overall, and 14 exact matches, and surpasses the frontier baseline on all three summary metrics. Across the 50 test items, it selects the frontier output 37 times, MT-DPO g0.4 sig 6 times, UNK-SFT 5 times, and synthesizes only 2 conservative repairs. Thus 48/50 outputs come from direct candidate selection, not free rewriting. The exact-match union of the three candidate sources is also 14/50, so the hybrid closes the exact oracle gap completely, although the oracle still retains headroom on chrF++ and overall score.

Method	Exact Match	Contam. Rate	Suffix Preserve	Len. Ratio
<i>Prompt-only</i>				
Dict prompt	0/50	0.00	3/27	1.83
<i>Frontier DeepSeek</i>	12/50	0.00	24/27	0.95
<i>Local learned</i>				
UNK-SFT	3/50	0.06	17/27	1.05
MT-SFT	5/50	0.02	19/27	1.10
Legacy DPO	2/50	0.18	18/27	1.88
MT-DPO g0.2 sig	5/50	0.00	19/27	1.74
MT-DPO g0.2 rob	4/50	0.14	23/27	1.58
MT-DPO g0.4 sig	5/50	0.04	22/27	1.12
MT-DPO g0.4 rob	5/50	0.04	22/27	1.26
<i>Hybrid workflow</i>				
<i>Catalog hybrid</i>	14/50	0.00	26/27	0.95
Human ref.	50/50	0.00	27/27	1.00

Table 2: Reference-aligned diagnostics derived from stored predictions. “Contam.” denotes mixed-script or artifact contamination. Bold marks the best local learned value in each column, excluding prompt-only baselines, the hybrid workflow, and the human reference.

System	Exact Match	chrF++	Overall
<i>Frontier DeepSeek</i>	12/50	28.54	2.80
2-way select hybrid	13/50	30.05	2.88
3-way catalog hybrid	14/50	34.87	3.02
Best-of-3 oracle	14/50	41.85	3.12

Table 3: From candidate complementarity to a real hybrid workflow. The 2-way hybrid arbitrates between the frontier system and MT-DPO g0.4 sig. The 3-way catalog hybrid adjudicates among the frontier system, UNK-SFT, and MT-DPO g0.4 sig, with optional minimal repair.

The new reproducibility baseline is useful for a different reason. A deterministic open selector over the same frontier/UNK-SFT/MT-DPO pool already recovers part of the workflow gain without closed adjudication: it reaches 32.62 chrF++, keeps zero contamination, matches the frontier’s 12/50 exact matches, and under the local Qwen judge ties the frontier baseline at 4.20 overall while still trailing the full catalog hybrid at 4.24. If one wants the best *single* model, the frontier baseline is still the right choice. The role of local training is instead to supply complementary historical-title hypotheses that a conservative adjudicator can exploit. This is visible in concrete cases where the hybrid restores 五部經 instead of the frontier normalization 五類經, or fixes the frontier truncation 番 to 番言金剛王乘根.

6.3 Reference-aware judging changes the ranking

The legacy LLM judge in the repository is reference-free, so we reran evaluation with a stricter judge that sees the Tangut input, the gold reference, and the candidate simultaneously.

Table 4 changes the ranking. Among single models, the frontier baseline remains the clear winner at 2.80 overall. Within the local learned family, the best overall score is shared by three multitask-base DPO variants at 2.22. But once we evaluate the actual workflow system, the catalog hybrid rises to 3.02 overall, 2.94 reference agreement, and 3.00 source faithfulness. The gains are

Method	Ref. Agr.	Src. Faith.	Title Style	Overall
<i>Prompt-only</i>				
Dict prompt	1.50	1.52	2.10	1.50
<i>Frontier DeepSeek</i>	2.78	2.82	4.48	2.80
<i>Local learned</i>				
UNK-SFT	1.78	1.84	4.36	1.82
MT-SFT	1.92	1.98	4.18	1.98
Legacy DPO	1.64	1.72	3.80	1.68
MT-DPO g0.2 sig	2.08	2.24	4.14	2.22
MT-DPO g0.2 rob	2.10	2.16	3.78	2.12
MT-DPO g0.4 sig	2.12	2.26	4.60	2.22
MT-DPO g0.4 rob	2.10	2.20	4.70	2.22
<i>Hybrid workflow</i>				
<i>Catalog hybrid</i>	2.94	3.00	4.72	3.02
Human ref.	5.00	5.00	5.00	5.00

Table 4: Reference-aware judge scores on a 1–5 scale. The judge sees the Tangut input, the gold reference, and the candidate output simultaneously. Bold marks the best local learned value in each column, excluding prompt-only baselines, the hybrid workflow, and the human reference.

Comparison ($B - A$)	Δ chrF++	Δ Overall	Δ Exact
Catalog hybrid – Frontier	+6.33 [2.10, 11.67]	+0.22 [0.00, 0.48]	+0.04 [0.00, 0.10]
Catalog hybrid – 2-way hybrid	+4.83 [1.40, 8.93]	+0.14 [−0.04, 0.34]	+0.02 [0.00, 0.06]
Local – MT-SFT	+4.02 [0.84, 7.80]	+0.24 [0.00, 0.46]	0.00 [0.00, 0.00]
Local – Frontier	+1.47 [−8.09, 10.29]	−0.58 [−1.04, −0.12]	−0.14 [−0.26, −0.04]
Legacy DPO – MT-SFT	−6.60 [−11.84, −1.73]	−0.30 [−0.64, 0.04]	−0.06 [−0.14, 0.00]

Table 5: Paired bootstrap comparisons with 5,000 resamples. “Overall” denotes mean reference-aware judge score, and “Exact” denotes exact-match rate.

concentrated where catalog conservatism matters most: on non-suffix titles the frontier baseline scores 2.39 overall and the hybrid 2.74, and on references of length ≤ 6 the frontier scores 2.94 and the hybrid 3.24. A local Qwen2.5-7B judge preserves the same frontier < hybrid < human ordering (4.20, 4.24, 4.90), reducing concern that the ranking is purely a GPT-family artifact.

6.4 Metric alignment, robustness, and filtered DPO

The `human_reference` run is an important anchor. The gold reference receives perfect chrF++ and perfect reference-aware judge scores, but only moderate lexical coverage and catastrophic perplexity. Lexical coverage therefore measures overlap with dictionary-derived semantics rather than agreement with the gold title, while perplexity penalizes compact title style and historical transliterations. These metrics remain useful diagnostics, but they should not drive model selection in this task.

The bootstrap results in Table 5 clarify which claims are stable. The local chrF++ gain from multitask SFT to strict gap-0.4 sigmoid DPO remains above zero under resampling, and the catalog hybrid’s chrF++ gain over the frontier baseline does as well. The judge-overall and exact-match gains are directionally positive but boundary-level on this 50-example test set. The overlap audit addresses a separate concern: there is zero exact source, target, or full-pair overlap across

Audit item	Value
Total preference pairs	3867
Duplicate chosen/rejected pairs	3
Non-finite reward entries	3
Mean reward gap	0.2396
Chosen closer to gold proxy	71.3%
Chosen closer on gap (0.05, 0.10]	62.3%
Chosen closer on gap (0.20, 0.40]	75.4%
Chosen closer on gap ≥ 0.40	81.5%
Pairs kept with gap ≥ 0.20	1996
Pairs kept with gap ≥ 0.30	1010
Pairs kept with gap ≥ 0.40	498

Table 6: Lightweight audit of the current DPO pair construction pipeline. Higher reward gaps correlate with cleaner preference labels under a simple gold-proxy check, which helps explain why strict filtering stabilizes the later multitask-base DPO runs.

train/dev/test; zero exact overlap between test references and the 50k synthetic multitask targets; and zero exact overlap between the frontier few-shot exemplars and the test references. This does not prove complete immunity from leakage, but it rules out the simplest exact-copy explanation.

The DPO audit explains why strict filtering helps. Across all 3,867 preference pairs, the reward-chosen candidate is closer to a simple gold proxy only 71.3% of the time, but this rises to 81.5% for gaps ≥ 0.40 . The high-gap filter is therefore isolating cleaner labels rather than acting as a cosmetic hyperparameter. Once DPO is moved onto the multitask base and filtered at 0.4, it stops behaving like a blanket negative result and becomes the strongest local model. The hybrid workflow does not erase this point; it depends on the local candidate pool being strong enough to contribute conservative alternatives in the first place.

7 Discussion and Limitations

The practical reading of the paper is simple. If a user can make only one model call, the DeepSeek-V3.2 frontier baseline is the best single model we measured. But if the target use case is scholar-facing short-title reconstruction rather than single-shot API convenience, the stronger claim is the workflow claim: a catalog-aware adjudicator over frontier and local candidates is better than frontier alone.

That distinction makes the contribution more general, not less. The local models matter not because they beat a frontier model head-to-head, but because they diversify the candidate pool precisely where fluent modern normalization is risky. This recipe should transfer beyond Tangut book titles: train structurally conservative local models, use them as hypothesis generators, and adjudicate with a title-aware reranker rather than a generic fluency prior.

The scope should also be explicit. The current dataset supports Tangut short-text translation, especially title-like strings, not general open-domain Tangut sentence translation. The strongest hybrid gains are concentrated on short and mid-length items, while long titles remain hard for every system. This is also why we do not foreground classical multilingual-transfer baselines: such methods are strongest when pretrained models already cover the source script or nearby related languages, and the present repository offers neither in a reproducible form. Under those conditions, lexicon-grounded synthesis and structured local adaptation are the more defensible comparison family. The same caveat applies to the synthetic corpus: it is a structural training proxy, not

realistic pseudo-Tangut. The new deterministic open reranker should likewise be read as a first reproducible hybrid baseline rather than the final word on open adjudication.

Two limitations matter most. First, the real dataset is extremely small, so the paper cannot claim stable behavior beyond this narrow domain. Second, both the adjudicator and the primary reference-aware judge currently rely on proprietary frontier models, and we do not yet provide a blinded human validation study for the judge. The local Qwen judge and deterministic open reranker reduce but do not remove this concern. To narrow the reproducibility gap, the repository should release the exact frontier prompting script, fixed few-shot exemplar IDs, adjudicator templates, per-item predictions, adjudicator outputs, and pair-filtering code. The most useful next experiments are therefore not “more DPO” in the abstract, but distilling the adjudicator into an open local reranker, validating the judge on a small expert-annotated subset, adding title-aware decoding controls, and testing transfer to other historical short-form translation settings.

8 Conclusion

The final message of this paper is now more precise and more useful. Strong frontier prompting does not fail on Tangut short-text translation; it is the strongest single-model baseline. Within the open/local regime, however, structured multitask supervision remains the right base recipe, and aggressively filtered multitask-base DPO produces the strongest local model. More importantly, those local models are not merely weaker copies of the frontier output. When their complementary historical-title hypotheses are combined through catalog-aware adjudication, the resulting workflow becomes the strongest non-reference system overall, reaching 34.87 chrF++, 3.02 reference-aware overall, and 14/50 exact matches.

This changes the practical lesson. For ultra-low-resource historical short-text translation, the right question is not simply whether local models can replace a frontier API. It is whether structured local training can produce conservative alternative hypotheses that improve a scholar-facing workflow. In this repository, the answer is yes. The broader methodological lesson is equally important: evaluation must be anchored to reference-aware criteria, because generic reference-free metrics can badly mis-rank compact historical titles and even penalize the gold reference itself.

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